(ADAM) CHEOL WOO KIM

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ACADEMIC APPOINTMENTS

Harvard University — Boston, MA
Postdoctoral Fellow in Computer Science — *Sep 2024 - Present*
- Advisor: Milind Tambe

EDUCATION

Massachusetts Institute of Technology — Cambridge, MA
PhD in Operations Research ; GPA: 5.0/5.0 — *Aug 2024*
- Advisor: Dimitris Bertsimas
- Thesis: *Predictive and Prescriptive Trees for Optimization and Control Problems*

Seoul National University School of Law — Seoul, Republic of Korea
J.D. Candidate (withdrew to pursue PhD at MIT) — *Feb 2019 - Jul 2019*

Seoul National University — Seoul, Republic of Korea
B.A. in Economics (Summa Cum Laude); Major GPA: 4.24/4.30 — *Feb 2019*
- Concentration in Mathematical Economics and Microeconomic Theory

RESEARCH INTERESTS

AI Alignment, Reinforcement Learning with Human Feedback, Human–Agent Interaction, ML-accelerated Optimization (Learning for Optimization), Intersection of Generative AI and Optimization

PAPERS

*: alphabetical or co-first author

Conference and Workshop Papers

- **Kim, C.W.***, Verma, S.*, Tec, M.* & Tambe, M. (2025). *Lightweight Robust Direct Preference Optimization*, NeurIPS 2025 Workshop: Reliable ML from Unreliable Data
- **Kim, C.W.***, Verma, S.*, Tec, M.* & Tambe, M. (2025). *Direct Preference Optimization with Preference Robustness*, AAAI Conference on Artificial Intelligence (AAAI)
- **Kim, C.W.***, Moondra, J.*, Verma, S., Pollack, M., Kong, L., Tambe, M., Gupta, S. (2025). *Navigating the Social Welfare Frontier: Portfolios for Multi-objective Reinforcement Learning*, International Conference on Machine Learning (ICML)
- Kong, L., Wang, H., Pan, Y., **Kim, C.W.**, Song, M., Nguyen, A., Wang, T., Xu, H., Tambe, M. (2025). *Robust Optimization with Diffusion Models for Green Security*, Uncertainty in Artificial Intelligence (UAI)

Journal Papers (Including Under Revision)

- Bertsimas, D.* & **Kim, C.W.*** (2024). *Optimal Control of Multiclass Fluid Queueing Networks: A Machine Learning Approach*, Minor Revision at Operations Research
- Bertsimas, D.*, **Kim, C.W.*** & Niño-Mora, J*. (2024). *Optimal Control of Fluid Restless Multi-armed Bandits: A Machine Learning Approach*, Major Revision at Machine Learning
- Bertsimas, D.* & **Kim, C.W.*** (2024). *A Machine Learning Approach to Two-stage Adaptive Robust Optimization*, European Journal of Operational Research
- Bertsimas, D.* & **Kim, C.W.*** (2023). *A Prescriptive Machine Learning Approach to Mixed Integer Convex Optimization*, INFORMS Journal on Computing

PROFESSIONAL EXPERIENCE

Massachusetts Institute of Technology Cambridge, MA
PhD Candidate *Sep 2019 - Aug 2024*

Core Thesis Topic

- Learning-accelerated algorithms for optimization and control.

Other Projects

- Developed machine learning models to predict patient flow at Beth Israel Deaconess Medical Center (a teaching hospital of Harvard Medical School), using medical data. The prediction models are fully integrated into their system.
- Collaborated with the OCP Group to develop a machine learning model to predict agricultural yield in Morocco combining image data and chemical composition data.
- Participated in a collaborative work to combat the spread of the Covid-19 pandemic: (https://www.covidanalytics.io/)

Microsoft Research Redmond, WA
Research Intern - Machine Learning and Optimization Group *May 2023 - Aug 2023*

- Research Topic: Theory and algorithms for combinatorial optimization under uncertainty.
- Mentors: Ishai Menache, Marco Molinaro, Konstantina Mellou

Permanent Mission of the Republic of Korea to the United Nations New York, NY
Research Intern - UN General Assembly Group *Jan 2014 - Apr 2014*

TEACHING EXPERIENCE

Developed course materials, led recitations, and held office hours for the following courses:

TA, The Analytics Edge (MIT 15.071) *Spring 2024*

- MBA course on statistics, machine learning, optimization, and data science.

TA, Optimization Methods (MIT 6.7200J/15.093J) *Fall 2022*

- Graduate course on linear, nonlinear, combinatorial, robust, and semi-definite optimization.

TA, Introduction to Mathematical Programming (MIT 6.251J/15.081J) *Fall 2021*

- PhD-level course on linear, nonlinear, discrete, combinatorial, robust, and semi-definite optimization.

TA, Robust Modeling, Optimization, and Computation (MIT 1.142J/15.094J) *Spring 2021*

- Graduate course on theory and applications of robust optimization.

REVIEWER SERVICES

Journal: Journal of Machine Learning Research, INFORMS Journal on Computing, INFORMS Journal on Optimization, European Journal of Operational Research, Transportation Science
Conference: AAAI 2026

AWARDS AND HONORS

Kwanjeong Educational Fellowship *Sep 2019 – Feb 2025*

- Merit-based PhD scholarship.

Eminence Scholarship *Aug 2017*
Department of Economics, Seoul National University Seoul, Republic of Korea

- Awarded to top 1% in the economics department.

SKILLS

- **Programming**: Python (PyTorch, Scikit-learn), Julia (JuMP, Gurobi, Mosek), R
- **Languages**: Korean (Native), English (Fluent)